

Trial Task

Order Flow Imbalance

May 3, 2025

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Introduction:

Order Flow Imbalance (OFI) serves as a powerful tool for quantifying supply and demand pressure within the limit order book. By capturing the net difference between buy and sell order flows, OFI can anticipate short-term price movements and reflect underlying market sentiment. This document is quite short. I highly recommend to spend more time on the python notebook, as it is much more informative and answers the same 3 questions anyway. Note that the dataset I downloaded on your dive only had market data for one stock (AAPL), which prevented me from running the true cross-sectional prediction model. To go around this problem, I would generate cointegrated stock data, so that it would mirror the behavior of stocks of the same industry. And I would then run the LASSO model on it. Or equally, if you have access to market data of stocks that are from the tech industry like apple, I would be happy to test this model and its predictive power.

The full implementation code can be found at: <https://github.com/charlesdezons/Cross-Impact-of-Order-Flow-imbalance-in-Equity-Markets>

1. What's the motivation behind measuring OFI at multiple depth levels of the order book?

Ans: The order book's entire depth provides information that the best level alone might miss. If a large hidden order is sitting at level 2 or 3, it won't affect best-level OFI until it hits the top, but a multi-level OFI measure would capture its presence earlier. In other words, the shape and imbalance of liquidity across multiple levels influence price dynamics, not just the first level. Empirically, incorporating order flow from deeper levels improves explanatory power for price changes. By summing or integrating OFIs up to 10 levels, we account for additional supply/demand that can move the price once the top of book is depleted.

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact??

Ans: In a multi-asset setting, a straightforward OLS regression including all stocks' OFIs as predictors would be high-dimensional and likely overfit, and it wouldn't automatically tell us which cross-asset effects are truly important. Lasso regression adds an L_1 penalty that performs variable selection, driving many coefficients to zero. This sparsity is desirable because we expect that a given stock's price is influenced by only a subset of other stocks (e.g. those in the same sector or with high correlations). Lasso thus yields a more interpretable model of cross-impact by highlighting only the significant cross-asset OFI predictors and ignoring spurious ones. It also helps mitigate multicollinearity and improves out-of-sample prediction by not fitting noise from dozens of irrelevant features.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

Ans: Trade volume is an unsigned measure – it counts the total shares traded but doesn't indicate the direction of imbalance. OFI, on the other hand, is a signed measure capturing the difference between buying and selling pressure. A million shares traded with equal buy and sell interest will produce no price move, but a net imbalance of even a smaller number of shares all in one direction (buying or selling) often moves the price. OFI encapsulates this net directional pressure. Moreover, OFI considers order book changes (including limit orders and cancellations) not just executed trades, so it reflects active demand and supply that may not have resulted in immediate trades but still affect prices. In short time intervals, price changes are mainly driven by order flow imbalance rather than sheer volume. Research has shown that models using OFI as a feature outperform those using raw volumes or past returns. Additionally, volume figures can be inflated by stale orders or symmetric trading, whereas OFI filters out the noise by focusing on the recent net order flow that actually tilts the price one way or the other. This makes OFI a superior predictor for short-term price movements.